

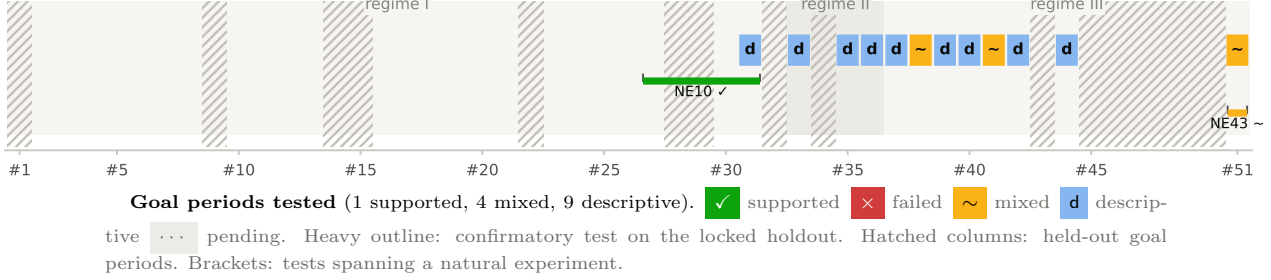
H35: The nudger is a measurably inefficient Maxwell demon

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 45% Faithfulness 2/5 Usefulness 2.5/5

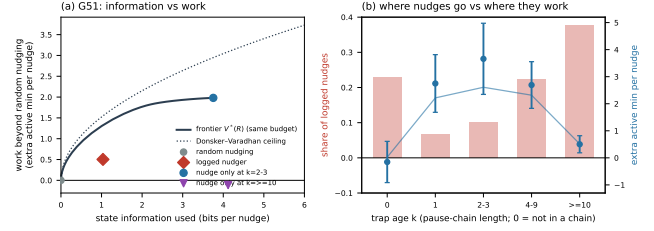
Bottom line. The nudger is a measurably inefficient Maxwell demon: it reads the right variable (trap age) but spends its nudges on the deepest, least responsive traps, and nudging once early in a pause chain would buy two to three times more work per nudge.

level: hypothesis · pipeline checks 86% · updated 2026-10-03



WHAT HAPPENED

The auto-nudger was treated as a feedback controller (a Maxwell demon). I measured how much agent-state information its nudges carry, how much activity they buy, and how close it comes to the best use of those bits, on 13 non-holdout periods plus two natural experiments (dated predictions; synthetic recovery checked first). In G51, the only well-powered period, a first nudge buys 1.45 [0.72, 2.20] extra active minutes using 1.4 bits of state. That is 38% of the work those bits could buy ($\eta_{SU} = 0.38 [-0.08, 0.68]$); only 15% of the bits were needed. Among pausing agents it does worse than random: 38% of nudges go to traps ≥ 10 pauses deep (0.5 min each, vs 2.2–2.6 at $k = 1-9$). Nudges add 0.5% of active minutes, invisible at switch-on (NE10) and at a silent stop on 08-20. Under the old 12-h pause default, nudges wake agents at any trap age. Holdout: not run.



Observables. (a) G51 in trap-age space: work beyond random nudging against state information used. The curve is the best any policy can do with that many bits (dotted: analytic ceiling). The logged nudger sits far below it; nudging only at $k = 2-3$ reaches the curve's end, and nudging only deep traps is worse than random. (b) The nudger's mass (bars) sits where the response per nudge (points) is lowest.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

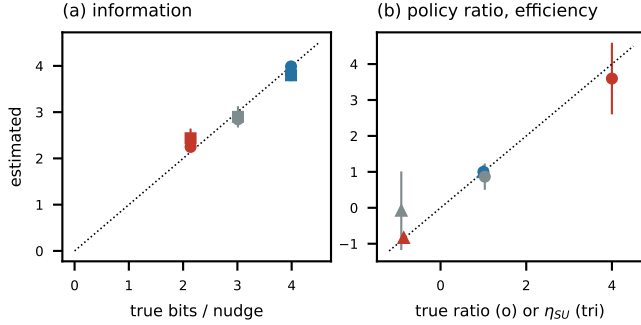
Models 04 (Kolchinsky–Wolpert) and 02. System X : an agent's state (idle time, pause timing, trap age $k = \text{pause-chain length}$). Demon: nudge $M \in \{0, 1\}$ at rate r . Work $g(x)$: extra active minutes in 30 min caused by a nudge in state x . Value of information ΔV : work beyond random nudging. Frontier $V^*(R)$: the most work any policy using R bits can buy. s : half-range of g (plays $k_B T$). p_k : probability of acting at the pause expiry.

$$\begin{aligned} \Delta V(\pi) &= \mathbb{E}[\pi g] - r \mathbb{E}[g], & I &= I(X; M) \\ V^*(R) &= \max_{I \leq R} \Delta V \Rightarrow \pi_\beta = \sigma(\logit r + \beta(g - \lambda)) \\ \eta_{SU} &= \Delta V_{\log} / V^*(I_{\log}), & \eta_{KW} &= R^*(\Delta V_{\log}) / I_{\log} \\ \Delta V &\leq s \sqrt{2rI} \quad (\text{Donsker-Varadhan}) \\ \logit p_k &= \alpha_i + \theta \ln k + (\gamma + \lambda \ln k) \mathbb{I}[\text{nudge}] \end{aligned}$$

Directly actionable, though the stakes are small. For an operator running an idle-nudger on LLM agents: nudge each pause chain once, early (the first or second re-pause), and only while the agent is paused. Stop re-nudging agents stuck for many pauses; they barely respond and need something else (a reset, a new task). In G51 that would double or triple the work per nudge at the same budget. The catch: the nudger buys only about 0.5% of agent activity, so even a perfect policy moves the swarm little unless you nudge far more. The thermodynamic framing works as accounting (frontier, ceiling, efficiency), not as physics. The policy result rests on one short-pause period and is untested on the holdout.

RESULTS VS PREDICTIONS

Prediction	Observed (G51 unless noted)	Verdict
P1 3–8 bits; agent-blind	1.42 bits/nudge; agent adds 0.43	partly
P2 work 0.5–3 min	1.45 [0.72, 2.20]; placebo clean	yes
P3 falls with trap age	$\lambda = -0.32 \pm 0.13$; G38 reversed	G51 only
P4 gates: $\Delta V > 0$, $\eta_{SU} \leq 0.5$	$\Delta V < 0$: worse than random	no (as worded)
P4 in trap-age space	$\eta_{SU} = 0.38$, $\eta_{KW} = 0.15$	bands met
P5 once early $\geq 1.25\times$	$\times 2.0$ – 2.2 escapes; $\times 2.3$ min	yes
P6 08-20 stop invisible	-1.4 pp [-4.8 , 2.0]; DiD $\approx$ drift	yes / inconcl.
P7 NE10 invisible	≤ 0.4 pp predicted; $+1.1$ pp seen	yes



Second figure. Synthetic validation (30 agents $\times$ 33 days, ~ 700 nudges, circles; G38-sized, squares). Information, the once-early vs logged ratio and η_{SU} are recovered (dotted: truth). Levels run 10–20% low; at ≤ 30 nudges the gate efficiencies are not identifiable.

GOAL PERIODS IN DETAIL

G31 d descriptive
G33 d descriptive
G35 d descriptive
G36 d descriptive
G37 d descriptive
G38 $\sim$ mixed
G39 d descriptive
G40 d descriptive
G41 $\sim$ mixed
G42 d descriptive
G44 d descriptive
G51 $\sim$ mixed
NE10 $\checkmark$ supported (accounting only; low information)
NE43 $\sim$ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Only G51 is powered, and it is the only short-pause period outside the holdout, so the policy result has no replication yet. Identification assumes the nudger acts on what we see; agent identity carries extra information, so it does not, fully. The efficiencies depend on the state space: about 0.7 coarse, 0.38 by trap age, below 0 among gates. The minute response per trap-age bin rests on 20–99 nudges. Placebos are clean only for the strict design. The 08-20 stop is undocumented and coincides with operator messages. Three post-hoc amendments are disclosed on the card.

ROUND 2

Run `confirm.py` on #47 and #50 (short pauses: C2–C4) and #45 (wake-up). Test the once-early rule as an actual intervention if operators allow. Add an LLM-state feature (Jev behavior labels) to see what the nudger reads beyond trap age.

No round-2 redirects yet.